

Rain Sensing Automatic Car Wiper

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Background & Problem

Today's car wipers work on the principle of manual switching. The problem is that manual intervention distracts the driver. We propose a system that automatically switches ON on detecting rain and stops when rain stops.



Project Objectives

- The goal is to automate the wiper system having no need for manual intervention.

1. Use rain sensor to detect rain.

2. Process signals using a microcontroller.

3. Drive a servo motor to simulate wiper action.

Operational Principle

- **Rain Detection Logic:**
 - The rain sensor uses water's conductivity to complete a circuit.
 - When dry, resistance is infinite. When wet, the circuit closes, sending a signal to the microcontroller.
- **Servo Motor Control (PWM):**
 - The servo is controlled by **Pulse Width Modulation (PWM)** signals.
 - A **1 ms pulse** moves the wiper to **0°**.
 - A **2 ms pulse** moves the wiper to **180°**.
- **Feedback Loop:**
 - A potentiometer inside the servo provides position feedback to ensure the wiper stays at the correct angle.